

Scalar theory: audited manuscript module

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1 Main-text candidate

1.1 From one deterministic edge to several stochastic diagnostics

On the scalar sampled quadratic $\ell_B(x) = h_B x^2/2 - \xi_B x$, SGD is exactly

$$X_{t+1} = A_{t+1}X_t + \eta\xi_{t+1}, \quad A_B = 1 - \eta h_B. \quad (1)$$

Two trajectories using the same sampled batches differ by $\Delta_{t+1} = A_{t+1}\Delta_t$. Write $M = |A_B|$, so moments always use the absolute multiplier, and define

$$m(p) = \mathbb{E}M^p, \quad \gamma = \mathbb{E} \log M, \quad H = \frac{1}{m(-1)}. \quad (2)$$

Here H is the harmonic gain; its inverse $m(-1)$ is the negative-moment coefficient. We only assign a finite inverse-moment estimate this meaning when the population inverse moment exists. For constant curvature, $m(p)^{1/p} = M$, $\gamma = \log M$, and $H = M$: the boundaries all coincide at $|1 - \eta h| = 1$. Curvature randomness separates them.

Proposition 1 (Scalar moment ordering). *Let $M > 0$, $\mathbb{E}|\log M| < \infty$, and suppose the displayed moments are finite. Then*

$$\log H \leq \gamma \leq \log m(1) \leq \frac{1}{2} \log m(2). \quad (3)$$

For iid multipliers and $\Delta_0 \neq 0$, $t^{-1} \log |\Delta_t/\Delta_0| \rightarrow \gamma$ almost surely and $\mathbb{E}|\Delta_t/\Delta_0|^p = m(p)^t$. Consequently $H > 1$ implies $\gamma > 0$. The converse need not hold.

The classical Kesten–Goldie result concerns the *affine* recursion (1): with $\gamma < 0$, nondegenerate additive forcing, and the usual renewal assumptions, its stationary upper tail has exponent $\kappa > 0$ solving $m(\kappa) = 1$ [Kesten, 1973, Goldie, 1991]. This is a baseline, not a theorem of post-expansion confinement. In particular, the nondegenerate Gaussian affine model used in Appendix A has no invariant probability when $\gamma > 0$. An additional return mechanism is needed.

1.2 Nonlinear return changes an upper-tail question into a central-mass question

The simplest exact return model is saturated amplitude:

$$R_{t+1} = \min\{r_*, M_{t+1}R_t\}, \quad 0 < R_t \leq r_*. \quad (4)$$

It separates local multiplicative motion from return at the outer scale. It is an explicit model, not an identity for a globally quadratic loss.

Theorem 2 (An inverse-moment law after local expansion). *Suppose $\mathbb{E}|\log M| < \infty$, $\gamma > 0$, $\log M$ is nonarithmetic, and $m(-\alpha) = 1$ for some $\alpha > 0$. Assume $m(-q)$ is finite in a neighborhood of α and $0 < -\mathbb{E}[M^{-\alpha} \log M] < \infty$. Then (4) has a unique invariant probability on $(0, r_*]$, and for some $C \in (0, \infty)$,*

$$\mathbb{P}_\pi(R \leq r) \sim C(r/r_*)^\alpha, \quad r \downarrow 0. \quad (5)$$

If $\log M$ additionally has a bounded continuous density with compact support, its radial density satisfies

$$p_R(r) \sim C\alpha r_*^{-\alpha} r^{\alpha-1}.$$

When the negative-moment curve is strictly convex and finite through 1 and α ,

$$H < 1 \iff \alpha < 1, \quad H = 1 \iff \alpha = 1, \quad H > 1 \iff \alpha > 1. \quad (6)$$

Thus positive local log growth can coexist with a central density cusp ($0 < \alpha < 1$); the harmonic crossing separates this cusp from central depletion ($\alpha > 1$) in this return model. For a reflection-symmetric signed coordinate, the same power occurs on each side. Central depletion is a local shape conclusion. Exactly two modes additionally requires control of the outer density, and arbitrary reinjection can reverse the conclusion (Appendix A.7).

Proof idea. The change of variable $W = \log(r_*/R)$ gives $W_{t+1} = \max\{0, W_t - \log M_{t+1}\}$. This is a reflected random walk with negative mean increment. Its invariant law is the supremum of the unreflected walk. Tilting the increment law by $M^{-\alpha}$ reverses its drift; the overshoot renewal theorem gives an exponential tail in W , hence the power (5) in R . The root therefore has the opposite sign from the Kesten upper-tail root.

1.3 An exact smooth SGD construction beyond both central boundaries

To prove that return can arise from a gradient update, rather than from an imposed cap, draw $h \in \{.95, 1.05\}$ equiprobably and $e \sim N(0, \sigma^2)$, independently, with $\sigma > 0$. Use

$$\ell(x; h, e) = \frac{1}{2}(\sqrt{h}x - e)^2 - \frac{.92}{2}\{x^2 - \log(1 + x^2)\}. \quad (7)$$

Every sampled loss is coercive. The population loss, up to a constant, is $L(x) = .04x^2 + .46 \log(1 + x^2)$, with unique minimizer zero but negative curvature on part of the line.

Theorem 3 (Smooth nonconvex SGD can return despite central expansion). *More generally, replace .92 in (7) by β and draw $h = \beta + \mu \pm s$ equiprobably, with $0 < s < \mu$ and $\beta > 8\mu$. At batch size one, every step $0 < \eta < 2/(\mu + s)$ has a unique attracting invariant probability with a smooth, strictly positive, even density and all finite polynomial moments. Its central Jacobian is $1 - \eta h$, whereas its outer slopes are $1 - \eta(\mu \pm s)$.*

For $(\beta, \mu, s, \eta) = (.92, .08, .05, 2.015)$, the central absolute gains are .91425, 1.11575, and

$$\gamma_0 = 0.009938 \dots > 0, \quad H_0 = 1.004999 \dots > 1, \quad \alpha_0 = 2.017427 \dots > 1, \quad (8)$$

where $\frac{1}{2}(.91425^{-\alpha_0} + 1.11575^{-\alpha_0}) = 1$. The outer slopes are .93955, .73805, both strictly contracting.

Proof idea. The exact update is $X^+ = A_\infty X - \eta\beta X/(1 + X^2) + \eta\sqrt{h}e$, with $A_\infty = 1 - \eta(h - \beta)$. The middle term is bounded and the outer slopes contract. A polynomial Foster drift and the positive Gaussian transition density give recurrence and uniqueness. The local derivative at the origin is expansive because the sampled curvature is much larger there. This theorem proves recurrence after the *central* Lyapunov and harmonic boundaries; it does not identify the tangent exponent along the invariant trajectory or infer its modes. These are separate empirical measurements below.

A Scalar theory: assumptions and complete derivations

A.1 Deterministic and iid quadratic products

For $\ell(x) = hx^2/2$, deterministic gradient descent satisfies $x_t = (1 - \eta h)^t x_0$. If $h > 0$, convergence to zero from every finite start occurs precisely when $0 < \eta h < 2$. At $\eta h = 2$, nonzero starts follow a two-cycle rather than converge, and at $\eta h > 2$ their magnitudes diverge. The mean, mean-square, logarithmic, and harmonic multiplier criteria coincide in this deterministic case. This calculation does not describe stationary fluctuations because there is no sampled forcing and no mechanism returning an expanding linear orbit to a bounded scale.

For iid batches, common-noise differences satisfy $\Delta_t/\Delta_0 = \prod_{j=1}^t A_j$, regardless of additive forcing. The logarithm is a sum of iid integrable variables, so the strong law gives the almost-sure exponent in Proposition 1. Independence also gives the exact p -moment identity, for every real p for which the moment is defined. For negative p , this identity describes inverse-distance growth; its contraction means repulsion from zero and must not be called contraction toward zero.

Jensen applied to $\exp(-\log M)$ gives $\log \mathbb{E}M^{-1} \geq -\mathbb{E}\log M$. Jensen applied to $\exp(\log M)$ gives $\mathbb{E}\log M \leq \log \mathbb{E}M$. Cauchy–Schwarz gives $\mathbb{E}M \leq (\mathbb{E}M^2)^{1/2}$. These are precisely (3). If M is nonconstant, the first Jensen inequality is strict whenever both sides are finite, so $H = 1$ then also implies $\gamma > 0$.

For example, let $M = 1/2$ with probability $1/5$ and $M = 2$ with probability $4/5$. Then $\gamma = (3/5)\log 2 > 0$, $m(-1) = 4/5$, $H = 5/4 > 1$, and $m(-2) = 1$. Changing the contracting probability to $2/5$ gives positive $\gamma = (1/5)\log 2$ while $m(-1) = 11/10 > 1$, hence $H < 1$. The separated thresholds therefore occur already for two-point curvature laws, not only for estimated high-dimensional diagnostics.

A.2 The affine baseline and why outer return matters

The following restricted setting suffices for an exact comparison and avoids unspoken nondegeneracy exceptions. Let (A_t, N_t) be iid, A_t take finitely many nonzero real values, and, conditionally on A_t , let N_t be centered Gaussian with variance in $[v_-, v_+] \subset (0, \infty)$. Multipliers and noise may be dependent. Consider

$$X_{t+1} = A_{t+1}X_t + r(X_t) + N_{t+1}, \quad \sup_x |r(x)| \leq R < \infty, \quad (9)$$

where r is continuous.

Proposition 4 (Outer logarithmic drift separates recurrence and escape). *If $\mathbb{E}\log|A| < 0$, (9) has a unique positive Harris invariant probability and converges geometrically in a weighted total-variation norm with weight $1 + |x|^p$ for some $0 < p \leq 1$. If $\mathbb{E}\log|A| > 0$, it has no invariant probability. No conclusion at equality is asserted.*

Proof. In the negative case, finiteness and nonvanishing of the multiplier support imply that $q(p) = \mathbb{E}|A|^p$ is differentiable at zero, with $q(0) = 1$ and $q'(0) < 0$. Choose $0 < p \leq 1$ with $q(p) < 1$. Subadditivity gives

$$P(1 + |x|^p) \leq q(p)(1 + |x|^p) + 1 - q(p) + R^p + \mathbb{E}|N|^p.$$

The transition density is a finite mixture of strictly positive Gaussian densities, continuous jointly in the starting and target state. On any two fixed compact intervals it has a positive lower bound. Every compact set is therefore small, the chain is Lebesgue irreducible and aperiodic, and the

geometric Foster–Harris theorem applies [Hairer and Mattingly, 2011]. To localize the drift, pick $q(p) < q' < 1$ and absorb its additive constant inside a sufficiently large level set of $1 + |x|^p$.

For the positive case write $a_j = |A_j|$, $P_n = \prod_{j=1}^n a_j$, and $d = \mathbb{E} \log a_1 > 0$. The reverse triangle inequality, iterated, gives

$$\frac{|X_n|}{P_n} \geq |X_0| - \sum_{j=1}^n \frac{R + |N_j|}{P_j}.$$

Fix $0 < c < d$. The strong law implies $\log P_n - cn \rightarrow \infty$, so $\inf_{n \geq 0} P_n e^{-cn} > 0$ almost surely. There is a deterministic $K > 0$ such that $E_K = \{P_n \geq K e^{cn} \text{ for all } n \geq 0\}$ has positive probability. Conditionally on the entire multiplier sequence, $\mathbb{E}(|N_j| \mid (A_k)_{k \geq 1}) \leq C = \sqrt{2v_+/\pi}$. On E_K , therefore,

$$\mathbb{E} \left[\sum_{j \geq 1} \frac{R + |N_j|}{P_j} \mid (A_k)_{k \geq 1} \right] \leq D := \frac{R + C}{K(e^c - 1)}.$$

For $|X_0| \geq 4D$, conditional Markov inequality gives probability at least $1/2$ that the weighted noise sum is at most $|X_0|/2$. Consequently the probability of $|X_n| \rightarrow \infty$ from every such large start is at least $\mathbb{P}(E_K)/2 > 0$. The everywhere positive one-step density reaches this region from every start with positive probability. An invariant law π would thus give positive stationary probability δ to escape. For any $M < \infty$, Fatou’s lemma and stationarity imply

$$\delta \leq \mathbb{E}_\pi \liminf_n \mathbf{1}_{\{|X_n| > M\}} \leq \liminf_n \mathbb{P}_\pi(|X_n| > M) = \pi(|x| > M).$$

Letting $M \rightarrow \infty$ contradicts that π is a probability. $\square$

For the affine case $r = 0$, negative logarithmic drift also gives the classical perpetuity representation $X \stackrel{d}{=} \sum_{n \geq 1} (A_1 \cdots A_{n-1}) N_n$. The products decay exponentially almost surely and Gaussian innovations grow subexponentially, so the series converges absolutely. Splitting off the first pair proves the distributional recursion. Under the additional Kesten–Goldie assumptions (a root $\mathbb{E}|A|^\kappa = 1$, $0 < \mathbb{E}|A|^\kappa \log |A| < \infty$, nonarithmetic $\log |A|$, the required κ -moment of the innovation, and no deterministic common fixed point), the stationary tails have power index κ . This last tail theorem is classical [Kesten, 1973, Goldie, 1991]; the theorem above proves the existence/nonexistence assertion needed here directly. Gaussian forcing excludes absorbing point masses and common deterministic fixed points.

A.3 Proof of the reflected-log central law

Let $Z_j = -\log M_j$, $S_n = \sum_{j=1}^n Z_j$, $S_0 = 0$, and $W = \log(r_*/R)$. Then

$$W_{t+1} = \max\{0, W_t + Z_{t+1}\}, \quad \mathbb{E} Z_1 = -\gamma < 0.$$

The strong law implies $S_n \rightarrow -\infty$, hence $W_* := \sup_{n \geq 0} S_n < \infty$ almost surely. Splitting the supremum at its first increment yields $W_* \stackrel{d}{=} \max\{0, Z_1 + W'_*\}$, with an independent copy W'_* . This proves invariance. To see uniqueness and attraction, couple two finite starting values with the same increments. The larger process reaches zero almost surely: before its first visit its value is its starting value plus S_n , which cannot stay positive forever. The smaller process is zero at that time by monotonicity, and the two processes then agree. Couple an arbitrary finite start to an independent stationary start this way. The coupling time is finite almost surely, so its tail tends to zero and total variation convergence follows.

Put $\phi(q) = \mathbb{E}e^{qZ} = m(-q)$. At the positive root $\phi(\alpha) = 1$, define the tilted increment law $F_\alpha(dz) = e^{\alpha z}F(dz)$. Its positive mean is $\mu_\alpha = \mathbb{E}[Ze^{\alpha Z}]$. For $T_u = \inf\{n \geq 1 : S_n > u\}$, the likelihood identity, first at each event $\{T_u = n\}$ and then summed over n , gives

$$\begin{aligned}\mathbb{P}(W_* > u) &= \sum_{n \geq 1} \mathbb{E}_\alpha[e^{-\alpha S_n} \mathbf{1}_{\{T_u = n\}}] \\ &= e^{-\alpha u} \mathbb{E}_\alpha e^{-\alpha O_u}, \quad O_u = S_{T_u} - u.\end{aligned}$$

Under the tilted law $T_u < \infty$ almost surely, since its drift is positive. The nonarithmetic overshoot renewal theorem, with the stated exponential-neighborhood and finite-mean assumptions, gives $O_u \Rightarrow O_\infty$, where O_∞ is finite and nonnegative. The bounded continuous function $e^{-\alpha o}$ yields $\mathbb{E}_\alpha e^{-\alpha O_u} \rightarrow C := \mathbb{E}_\alpha e^{-\alpha O_\infty} \in (0, 1]$. This is the classical Cramér–Lundberg asymptotic; its random-walk renewal formulation is also described by [Blanchet and Glynn \[2006\]](#). Substitution of $u = \log(r_*/r)$ proves the small-ball law. Strict versus weak inequalities have the same asymptotic by squeezing between radii r and $(1 + \epsilon)r$ and then letting $\epsilon \downarrow 0$.

Here is a proof of the concrete density condition in the theorem. Suppose $Z = -\log M$ has a bounded continuous density f , supported in $[-b_-, b_+]$, with $b_+, b_- > 0$. Let τ_+ be its first strictly positive ladder epoch and define the defective ladder-height measure $G(D) = \mathbb{P}(\tau_+ < \infty, S_{\tau_+} \in D)$ on $(0, b_+]$. Write $p = G((0, b_+])$. Since $S_n \rightarrow -\infty$, the walk has finitely many strict upper records almost surely. The strong Markov property would produce infinitely many records if $p = 1$, so $p < 1$. Decomposing the supremum by its number of records gives

$$\mathcal{L}(W_*) = (1 - p) \sum_{n \geq 0} G^{*n}.$$

The ladder measure has density

$$g_G(x) = \mathbf{1}_{\{0 < x \leq b_+\}} \sum_{n \geq 0} \mathbb{E}[\mathbf{1}_{\{\tau_+ > n\}} f(x - S_n)].$$

The summand is zero unless $-b_+ \leq S_n \leq 0$. A bounded-increment random walk with negative mean has finite expected occupation of this compact interval: for large n , the event $S_n \geq -b_+$ is an exponentially unlikely deviation from its negative mean, by Hoeffding’s inequality. Thus g_G is bounded. Dominated convergence using the same finite occupation measure makes it continuous on $(0, b_+)$, with finite one-sided endpoint limits. It is consequently directly Riemann integrable and compactly supported.

Exponential tilting at the stopped ladder epoch gives $\widehat{G}(dx) = e^{\alpha x} G(dx)$, which is a probability: under the tilted positive-drift walk the first positive ladder epoch is finite almost surely. Its mean $\widehat{\mu} = \int x \widehat{G}(dx)$ lies in $(0, b_+]$, and it has a bounded, directly Riemann integrable density $\widehat{g}$. Let $\widehat{U} = \sum_{n \geq 0} \widehat{G}^{*n}$. The positive-increment key renewal theorem applied to $\widehat{g}$ gives

$$\widehat{u}(w) := \int \widehat{g}(w - y) \widehat{U}(dy) \longrightarrow \frac{1}{\widehat{\mu}}.$$

This convolution is a density of $\widehat{U} - \delta_0$, since $\widehat{U} = \delta_0 + \widehat{G} * \widehat{U}$. Undoing the tilt in the supremum decomposition therefore gives a density on $(0, \infty)$,

$$g_{W_*}(w) = (1 - p) e^{-\alpha w} \widehat{u}(w) \sim \frac{1 - p}{\widehat{\mu}} e^{-\alpha w}.$$

Its integral tail identifies the earlier constant as $C = (1-p)/(\alpha\hat{\mu})$. Finally $p_R(r) = g_{W_*}(\log(r_*/r))/r$, proving the density assertion. The atom $1-p$ at $W_* = 0$, equivalently at $R = r_*$, is retained. This argument derives the local density from the positive-increment renewal theorem and does not differentiate a CDF asymptotic.

For the harmonic comparison, let $\psi(q) = \log \phi(q)$. It is strictly convex for a nonconstant gain, with $\psi(0) = 0$ and $\psi'(0) = -\gamma < 0$. Its unique positive zero is α . It is negative on $(0, \alpha)$ and positive beyond α within its finite domain. Since $\psi(1) = -\log H$, this proves (6). Existence of the root is a substantive assumption: if $M > 1$ almost surely then no positive inverse root exists.

A.4 A solvable return model with an exact density

Proposition 5 (Exponential log excursions). *Let $U \sim \text{Exp}(b)$, $V \sim \text{Exp}(c)$ be independent, $b > c > 0$, and $M = e^{V-U}$. In (4), set $\alpha = b - c$ and $\rho = c/b$. Then*

$$\pi_R = (1 - \rho)\delta_{r_*} + \rho \frac{\alpha}{r_*} \left(\frac{r}{r_*} \right)^{\alpha-1} \mathbf{1}_{(0, r_*)}(r) dr. \quad (10)$$

The local logarithmic drift is $1/c - 1/b > 0$ and $m(-q) = bc/[(b-q)(c+q)]$ for $0 \leq q < b$. Thus $m(-\alpha) = 1$, and, when $b > 1$, $H > 1$ precisely when $\alpha > 1$.

Proof. For the reflected coordinate, propose $W \sim (1 - \rho)\delta_0 + \rho \text{Exp}(\alpha)$, independent of fresh U, V . Its Laplace transform is

$$\mathbb{E}e^{-sW} = 1 - \rho + \rho \frac{\alpha}{\alpha + s} = \frac{\alpha(b + s)}{b(\alpha + s)}.$$

Multiplying by the Laplace transform $b/(b + s)$ of U shows $W + U \sim \text{Exp}(\alpha)$. Therefore, for $w \geq 0$,

$$\mathbb{P}((W + U - V)_+ > w) = e^{-\alpha w} \mathbb{E}e^{-\alpha V} = \rho e^{-\alpha w}.$$

This is the proposed invariant law; uniqueness follows from the preceding coupling argument. Transforming back gives (10). Independence gives the displayed moment formula, whose nonzero root is $b - c$. Finally $H = (b - 1)(c + 1)/(bc)$, so $H - 1 = (\alpha - 1)/(bc)$. $\square$

The boundary atom is real and must be retained in plots or simulations. A fair independent sign gives a symmetric invariant coordinate with half the mass on each side, but its two boundary atoms are not a theorem about two smooth interior modes in neural-network coordinates.

A.5 Finite-mean reinjection: a useful generalization

The hard cap can be replaced by independent excursions. The following version uses bounded increments and entries to make the renewal averaging step transparent. Suppose $Y = \log M$ has a bounded continuous density with bounded support, $\mathbb{E}Y > 0$, $\mathbb{P}(Y < 0) > 0$, and $\mathbb{E}e^{-\alpha Y} = 1$ for $\alpha > 0$. Let the independent entry Z lie in $[-L, -\ell]$, with $0 < \ell \leq L < \infty$. Run $Q_n = Z + \sum_{j=1}^n Y_j$ until $\tau = \inf\{n \geq 0 : Q_n \geq 0\}$; discard the exit and restart at a fresh independent Z . Then

$$\pi_Q(D) = \frac{\mathbb{E} \sum_{n=0}^{\tau-1} \mathbf{1}_{\{Q_n \in D\}}}{\mathbb{E}\tau} \quad (11)$$

is invariant. To see $\mathbb{E}\tau < \infty$, write $\mu = \mathbb{E}Y > 0$. Since $\{\tau > n\}$ implies $\sum_{j=1}^n Y_j < L$, bounded-increment Hoeffding bounds make the tail summable for $n > 2L/\mu$. For bounded test functions, the expected one-step change summed over a cycle telescopes to the next fresh entry minus the present entry; its expectation is zero. This proves (11) is invariant.

Let $m_\alpha = -\mathbb{E}[Ye^{-\alpha Y}] > 0$ and $D_\alpha = \mathbb{E}[e^{-\alpha Z} - e^{-\alpha Q_\tau}] > 0$. For every fixed $h > 0$, the renewal calculation gives

$$\pi_Q((x, x+h]) \sim \frac{D_\alpha}{\alpha m_\alpha \mathbb{E}\tau} (e^{\alpha h} - 1)e^{\alpha x}, \quad x \rightarrow -\infty. \quad (12)$$

Here is the calculation. For the increment law F , write $U = \sum_{n \geq 0} F^{*n}$. Tilting by $e^{-\alpha y}$ yields a probability F_α with negative drift $-m_\alpha$ and $U(dt) = e^{\alpha t} U_\alpha(dt)$. The nonarithmetic random-walk renewal theorem, applied on fixed intervals and then to continuous test functions on those intervals, gives

$$U_z((x, x+h]) \sim e^{-\alpha z} \frac{e^{\alpha h} - 1}{\alpha m_\alpha} e^{\alpha x}.$$

This convergence is uniform over z in a compact set. One can obtain uniformity from convergence on a finite grid of interval endpoints and monotone upper/lower interval approximations. The original entry is in a compact interval, and the exit overshoot lies in $[0, y_+]$, where y_+ bounds the positive increments. Thus the averaging is valid without any unbounded-entry interchange. The strong Markov property gives the exact killed-potential identity

$$U_Z(D) = \mathbb{E} \sum_{n < \tau} \mathbf{1}_{\{Q_n \in D\}} + \mathbb{E} U_{Q_\tau}(D).$$

Subtract the two uniformly valid asymptotics and divide by $\mathbb{E}\tau$ to obtain (12). Bounded local renewal masses give a uniform exponential bound for the fixed-width intervals; a geometric-series domination then permits summing toward $-\infty$. Hence for $R = e^Q$,

$$\mathbb{P}_\pi(R \leq r) \sim \frac{D_\alpha}{\alpha m_\alpha \mathbb{E}\tau} r^\alpha.$$

The bounded continuous increment density also verifies the density version here directly. The tilted unrestricted potential satisfies $U_\alpha = \delta_0 + F_\alpha * U_\alpha$. Its absolutely continuous part therefore has the continuous density $u_\alpha(x) = \int f_\alpha(x-y) U_\alpha(dy)$. The fixed-interval renewal theorem gives vague convergence of $U_\alpha(x+dt)$ to dt/m_α as $x \rightarrow -\infty$. Because f_α is continuous and compactly supported, this convolution converges to $1/m_\alpha$. The convergence is uniform under bounded shifts. At sufficiently negative locations the compactly supported entry and exit measures have no atoms; the killed-potential subtraction therefore has a density there. Undoing the tilt and subtracting its entry and exit coefficients proves $p_R(r) \sim D_\alpha r^{\alpha-1}/(m_\alpha \mathbb{E}\tau)$. This result proves an excursion law with specified return. Applying it to a nonlinear SGD path requires verifying its return and approximation hypotheses, including the distribution of entry near zero.

A.6 Proof of the smooth nonlinear SGD example

For the general parameters in Theorem 3, expansion of the sampled loss gives

$$\ell(x; h, e) = \frac{1}{2}(h - \beta)x^2 + \frac{\beta}{2} \log(1 + x^2) - \sqrt{h}ex + \frac{1}{2}e^2.$$

Because $h - \beta \geq \mu - s > 0$, every realized loss is coercive. Taking expectations gives $L(x) = \mu x^2/2 + (\beta/2) \log(1 + x^2) + C$. Its derivative is $x[\mu + \beta/(1 + x^2)]$, so zero is the unique minimizer. Its second derivative is

$$L''(x) = \mu + \beta \frac{1 - x^2}{(1 + x^2)^2}.$$

The fraction attains its minimum $-1/8$ at $x^2 = 3$; thus $\min L'' = \mu - \beta/8 < 0$. Differentiation of the sampled loss gives the exact step

$$X^+ = (1 - \eta(h - \beta))X - \frac{\eta\beta X}{1 + X^2} + \eta\sqrt{h}e.$$

The remainder has absolute value at most $\eta\beta/2$. The assumptions on η imply $a_* := \max\{|1 - \eta(\mu - s)|, |1 - \eta(\mu + s)|\} < 1$. For any $p > 0$, choose $\delta > 0$ so $q = (1 + \delta)a_*^p < 1$. The elementary bound

$$|u + v|^p \leq (1 + \delta)|u|^p + C_{p,\delta}|v|^p$$

(or subadditivity when $p \leq 1$) gives $P|x|^p \leq q|x|^p + C_p$, since the remainder is bounded and the Gaussian noise has all moments. At $p = 1$ this drift, combined with the Gaussian compact minorization in the proof of Proposition 4, gives recurrence, uniqueness, and convergence. This direct argument also covers a zero outer slope, which was excluded from the logarithmic-sign proposition. Iteration from any fixed finite start bounds $\sup_n \mathbb{E}|X_n|^p$. Convergence in law and the Portmanteau theorem give $\int |x|^p \pi(dx) < \infty$, without assuming this moment when integrating the drift.

The conditional noise variances are $\eta^2\sigma^2h$, bounded above and away from zero. Invariance writes the density as a mixture of translated Gaussian densities. Every Gaussian derivative of fixed order is uniformly bounded over its translation and this compact variance interval. Differentiating under the invariant mixture shows that the density is bounded and smooth; positivity is immediate. Reflection equivariance and uniqueness give evenness. Finally the exact sampled Jacobian at zero is $1 - \eta h$. At the numerical instance in the theorem, the signed gains are $-.91425, -1.11575$, whose product is greater than one. Their harmonic mean is greater than one by direct rational arithmetic. Strict convexity of $q \mapsto \mathbb{E}|1 - \eta h|^{-q}$, its negative derivative at zero, and its divergence as $q \rightarrow \infty$ (from the contracting atom) give a unique positive root. Evaluating this finite two-term function gives (8); the signs are exact and do not depend on the displayed decimal rounding.

Increasing β while holding (μ, s, η) fixed leaves the outer slopes unchanged and makes both central gains grow without bound. Every fixed finite β still has a bounded remainder and finite Gaussian moments. Consequently recurrence and all finite polynomial moments can coexist with arbitrarily large central logarithmic and harmonic gains, although a contracting central atom and hence a finite inverse root eventually disappear.

A contrasting occupied inverse moment at a larger step. The same fixed loss at $\eta = 8$ has outer slopes $.76, -.04$ and central Jacobians $-6.6, -7.4$, so all recurrence conclusions remain valid and $H_0 = 6.977142857\dots > 1$.

The Jacobian at an occupied state is instead

$$D_h(x) = 1 - 8(h - .92) - 7.36 \frac{1 - x^2}{(1 + x^2)^2}.$$

It is state dependent. In fact its simple zeros, together with the strictly positive invariant density, imply $\mathbb{E}_{\pi,h}|D_h(X)|^{-1} = \infty$: the integrand is comparable to $1/|x - x_c|$ near any simple zero x_c . Thus the occupied harmonic gain can be zero while the central frozen harmonic gain exceeds one. A finite Monte Carlo inverse moment need not detect this divergence. This reinforces why all scalar directions and locations must be identified in an empirical threshold claim.

When central depletion gives off-center modes. Suppose a signed invariant law has a continuous even density f on $\mathbb{R}$, $f(0) = 0$, $f(x) \rightarrow 0$ as $|x| \rightarrow \infty$, and positive total mass. Some $x_0 > 0$ has $f(x_0) > 0$. The limit at zero and infinity confines the supremum on $[0, \infty)$ to a compact subinterval of $(0, \infty)$, where continuity attains it. Its negative reflection is another maximizer. Thus there are at least two off-center modes. This does not exclude additional modes or a plateau. Exactly two requires, for example, a unique maximum on each half-line. Moreover, at fixed nonzero additive Gaussian noise the density in the theorem is positive at zero, so an ideal multiplicative depletion law cannot be transferred literally down to zero without a small-noise argument.

A.7 A counterexample to a universal harmonic-to-bimodality claim

On $(0, 1)$, take $M = 1/2$ with probability $1/5$ and $M = 2$ otherwise. If $MR < 1$, set $R' = MR$; otherwise independently restart from density

$$g(r) = \begin{cases} (24 + 21r^2)/41, & 0 < r < 1/2, \\ (72 - 27r^2)/41, & 1/2 \leq r < 1. \end{cases}$$

The multiplier has $\gamma = (3/5)\log 2 > 0$, $H = 5/4 > 1$, and inverse root $\alpha = 2$. Nevertheless the exact invariant density is

$$f(r) = \frac{12}{11}(1 - r^2/4), \quad 0 < r < 1,$$

which decreases from the origin. Both displayed functions integrate to one. The killed transfer operator is

$$Tf(r) = \frac{2}{5}f(2r)\mathbf{1}_{\{r < 1/2\}} + \frac{2}{5}f(r/2).$$

Direct substitution on the two half-intervals gives $f - Tf = (41/110)g$. Under f , the reset probability is $(4/5)\int_{1/2}^1 f(r)dr = 41/110$, so $f = Tf + (41/110)g$ proves invariance. Adding a reflection-symmetric sign yields a smooth central maximum. The return law places substantial mass arbitrarily near zero and does not satisfy the inverse entry-moment condition of the more general regenerative theorem. The multiplier alone cannot distinguish this counterexample from a regular-return model with the same α .

A.8 Exactly two modes in the confined diffusion normal form

For clarity, one setting where mode count can be proved completely is the *diffusion* normal form

$$dY = (aY - cY^3)dt + \sqrt{\sigma^2 + \nu^2 Y^2} dW, \quad c, \sigma, \nu > 0.$$

The quartic inward drift and nondegenerate diffusion give a nonexplosive positive recurrent scalar process. Solving its zero-current equation $(Dp)' = 2(ay - cy^3)p$, where $D = \sigma^2 + \nu^2 y^2$, gives

$$p(y) = C(\sigma^2 + \nu^2 y^2)^{a/\nu^2 + c\sigma^2/\nu^4 - 1} e^{-cy^2/\nu^2}.$$

It is normalizable, and direct differentiation yields

$$(\log p)'(y) = \frac{2y[(a - \nu^2) - cy^2]}{\sigma^2 + \nu^2 y^2}.$$

There is one mode at zero for $a \leq \nu^2$, and exactly two modes at $\pm\sqrt{(a - \nu^2)/c}$ for $a > \nu^2$. The zero-additive-noise local logarithmic exponent is $a - \nu^2/2$. When $a > \nu^2/2$, its nontrivial zero-noise

invariant density on the punctured line has central exponent $\alpha = 2a/\nu^2 - 1$, so the mode transition occurs at $\alpha = 1$. This is a diffusion theorem; fixed-step SGD requires its own confinement and approximation argument. In particular, a raw cubic explicit update with nondegenerate Gaussian innovations can escape and does not inherit this invariant law merely because its local Taylor expansion matches the diffusion drift.

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